GRAPH NEURAL NETWORKS · COURSE PROJECT

Protein Interface Prediction with Graph Convolutional Networks

Which residues of two proteins will touch when they bind?
A PyTorch implementation of the graph-convolution method, trained and evaluated on the Docking Benchmark, with an edge-aware extension and a verified evaluation.

Base paper: Fout, Byrd, Shariat & Ben-Hur, NeurIPS 2017 · Data: Docking Benchmark 5 / DB5-Plus

Which paper, and why this one

Paper chosen

Protein Interface Prediction using Graph Convolutional Networks
Fout et al., NeurIPS 2017

Why it fits the assigned topic

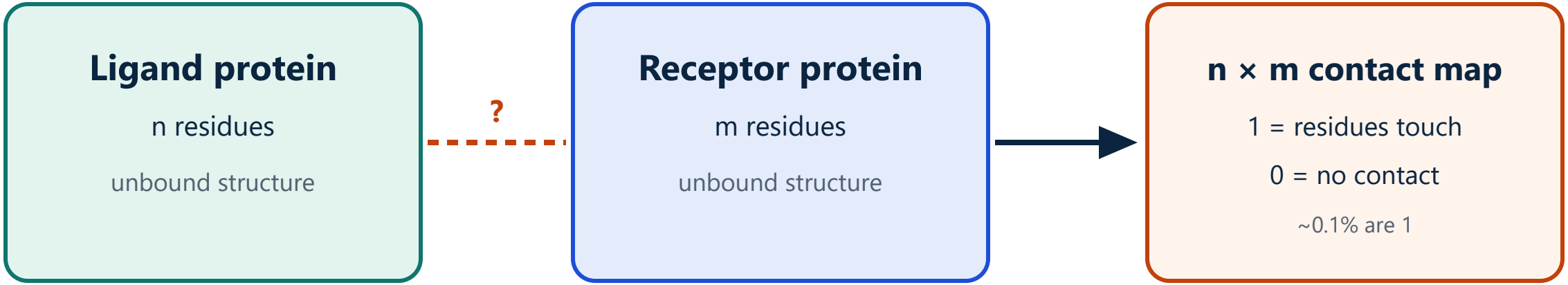
- Topic row: **Protein Interface Prediction → GCN**.
- The paper learns **ligand** and **receptor** residue representations separately, then **merges them** for pairwise classification: exactly the task description.
- Releases code and benchmark data, so results can be checked.

Why the problem matters

- Interfaces decide which proteins bind: key to **drug discovery** (where a drug must bind or block).
- Experimental structures of complexes are slow and expensive; a model that predicts contact residues narrows the search.
- Graphs are the natural fit: a protein is a set of residues linked by spatial neighbourhood, not a grid or a sequence.

Goal: given two *separate* proteins, score every (ligand residue, receptor residue) pair for contact.

What exactly is predicted



- **Input:** two proteins in their **unbound** (separate) state: the realistic case, since the bound shape is unknown beforehand.
- **Label:** from the **bound** complex: a residue pair is a contact if any non-hydrogen atoms are within **6 Å**.
- **Output:** a score for every cross-protein residue pair. It is a **residue-pair classification** task, not folding or docking.
- **Challenge:** extreme imbalance: ~1 contact per 1,000 pairs.

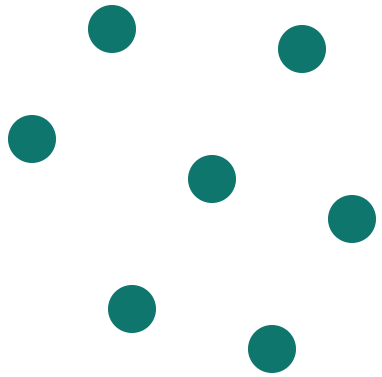
From a 3-D protein to a residue graph

① Protein chain



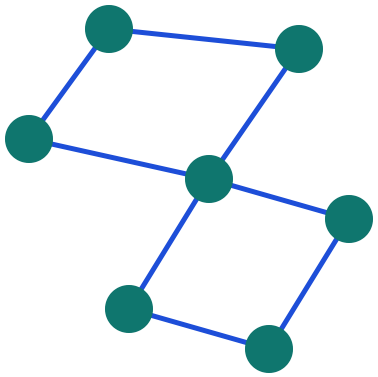
backbone of amino acids

② Residues = nodes



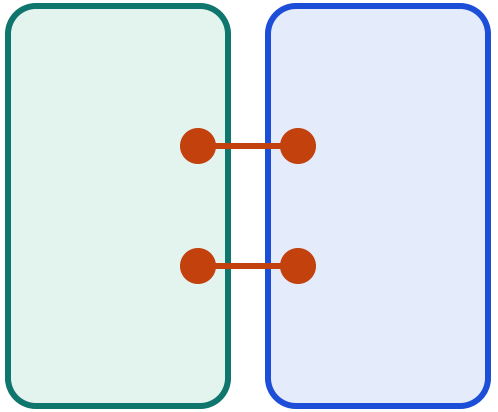
each node: features of that residue

③ 20 nearest residues = edges



edge = spatially close (not just sequence-adjacent)

④ Label from bound state



atoms within 6 Å ⇒ contact = 1

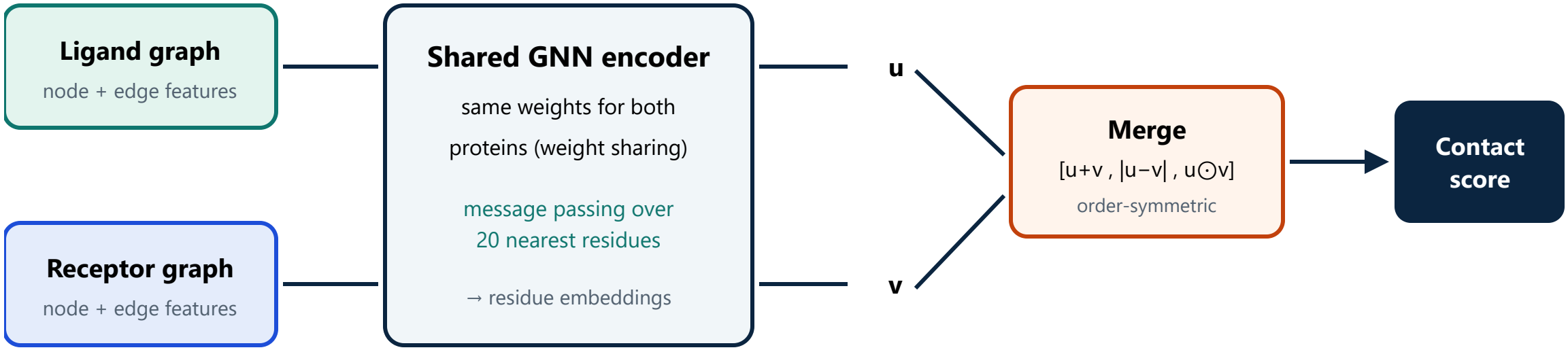
Unbound state (model input)

Proteins are separate. Residue features and neighbourhoods are computed here: no knowledge of the partner pose.

Bound state (labels only)

Only the 6 Å contact labels come from the docked complex. Cross-protein distances are **never** given as input.

Architecture: two shared encoders, one pair classifier



- **Why shared weights:** a residue is described the same way whether it is in the ligand or receptor, and the model is invariant to which partner is called ligand.
- **Why the symmetric merge:** sum, absolute difference and product keep the pair score independent of partner order.
- **Matched baseline:** identical model with message passing removed (MLP), so any gain is attributable to graph context.

The graph-convolution layer

Paper's node-average operator (Eq. 1)

$$h_i' = \mathrm{ReLU}\left(W_C\, h_i + W_N \cdot \mathrm{mean}_{j \in N(i)}\, h_j + b \right)$$

h_i: residue i · N(i): its 20 nearest residues · W_C, W_N: separate learned weights for centre and neighbours.

Intuition

- A residue's interface role depends on its **surface patch**, not just itself.
- Each layer lets information travel one neighbourhood further; two layers see a wider patch.

Our Kaggle GNN (extension)

- Neighbour messages are **weighted by edge features** (distance radial-basis + sequence gap): the paper's "node-and-edge" idea.
- **3 residual layers**, width 128, dropout 0.25.
- Pair head: sum / |difference| / product.
- Small auxiliary single-residue interface loss.

Fair baseline

MLP: same width, head, features and loss: **no neighbour messages**.

Dataset and splits (no leakage)

Partition	Complexes	Used for
Official training	140	Learn weights + scaler
Checkpoint selection	18	Pick epoch (mean AP)
Threshold calibration	17	Fix F1 threshold
Official test	**55**	**Final numbers, touched once**

The 18/17 split is a fixed seed-224 split of the 35 official validation complexes.

Test set

5,224,053

candidate pairs · **4,891 contacts** (0.094 %)
No negative downsampling at validation or test.

- Docking Benchmark 5 (DB5-Plus release); unbound features, bound labels.
- Median imputation + scaling fitted on training proteins only.
- Features: **218 node**, **17 edge** (incl. missing-value flags).
- Archive checksum verified.

How the models were trained

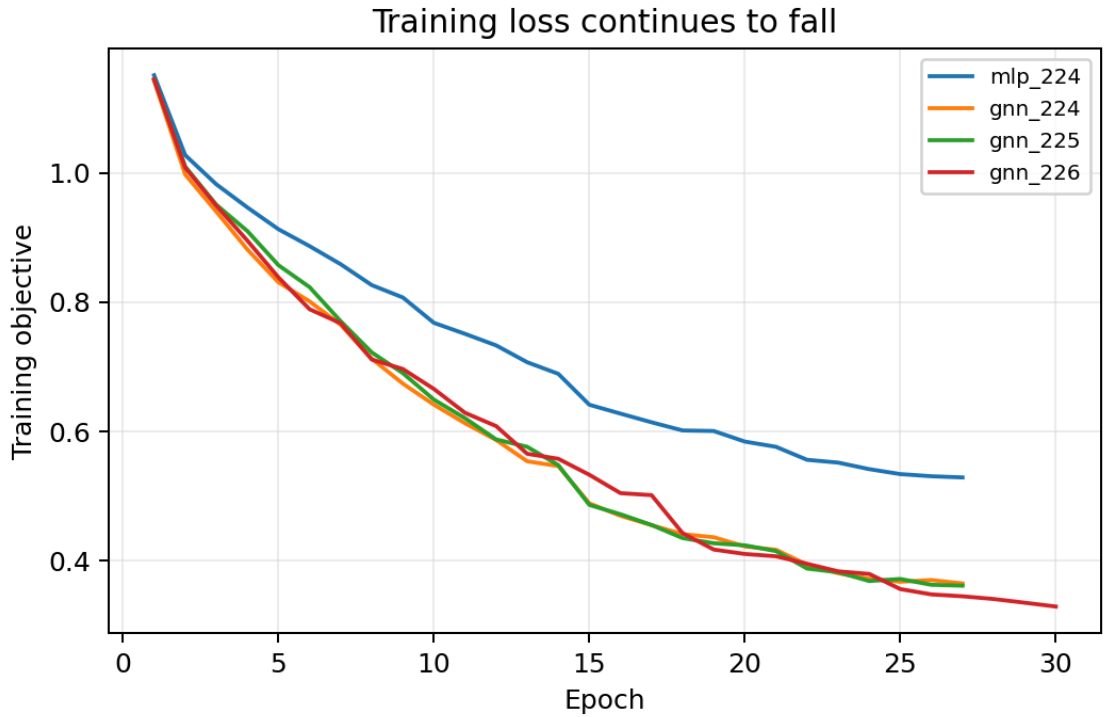
Setting	Value
Hardware	Kaggle 2 × Tesla T4, 385.4 s
Optimiser	AdamW, lr 3e-4, wd 1e-3
Regularisation	dropout 0.25, grad clip 5
Negatives	10 per positive, resampled each epoch
Schedule	LR drops on validation plateau
Stopping	≤150 epochs, patience 20
Runs	MLP ×1, GNN ×3 seeds

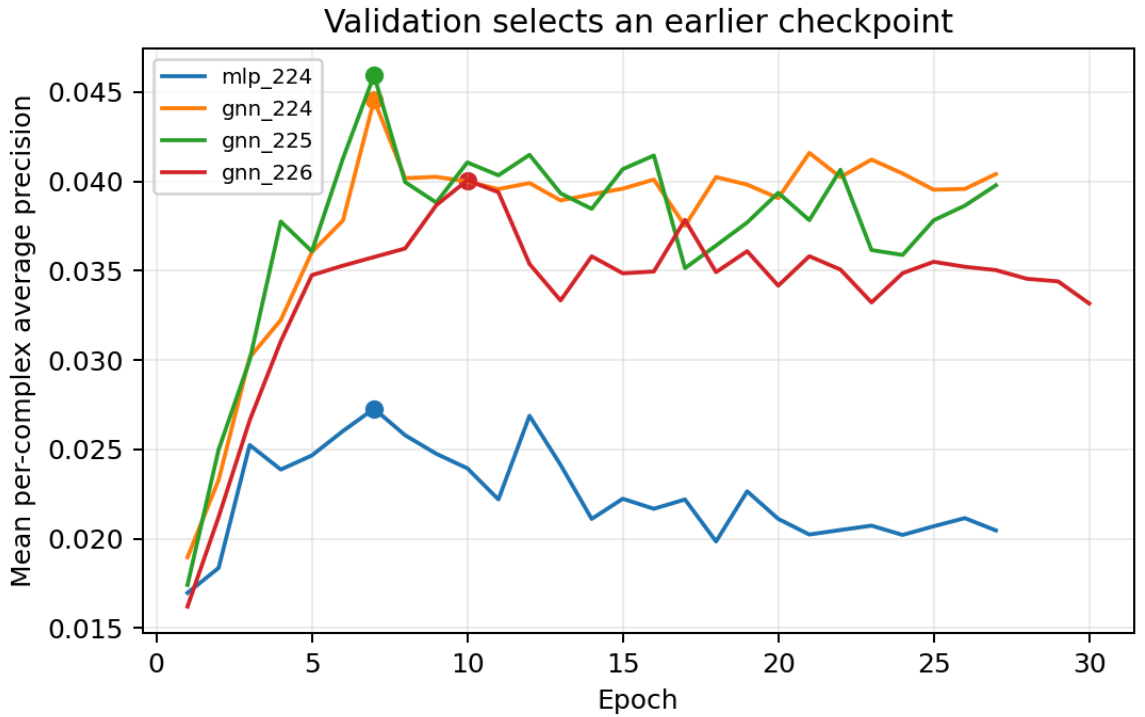
Process

1. Fit scaler on training proteins.
2. Each epoch: sample positives + 10 negatives, balanced loss.
3. Score validation complexes with **mean per-complex AP**; keep best epoch.
4. Calibrate F1 threshold on 17 held-back complexes.
5. Evaluate once on 55 test complexes.
6. Ensemble the 3 GNN seeds (rule declared before testing).

> Best epochs were 7–10: validation AP peaked early while training loss kept falling: classic overfitting caught by early stopping.

Validation picks the checkpoint, not the loss





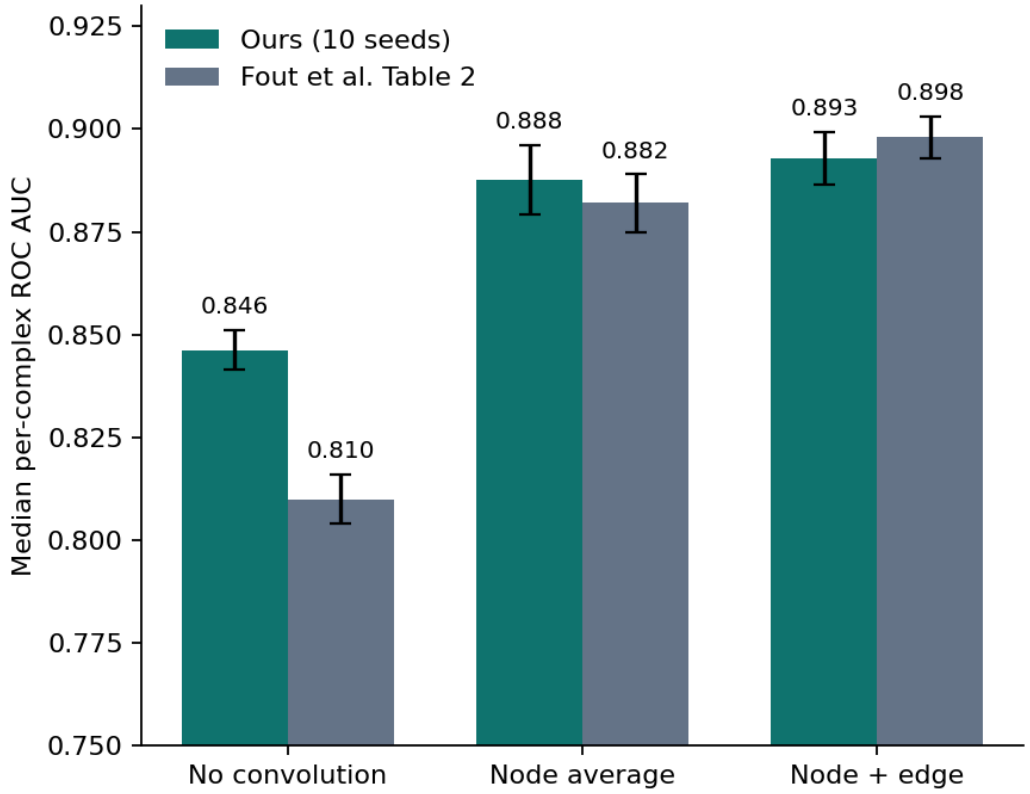
Training loss keeps decreasing, but validation average precision stops improving after epoch 7–10, so the saved weights come from those epochs, not the last epoch.

Our results agree with the published benchmark

2 layers · 55 test · 10 seeds	Our results	Published (Fout et al.)
No convolution	0.846 ± 0.005	0.810 ± 0.006
Node average (Eq. 1)	0.888 ± 0.008	0.882 ± 0.007
Node + edge average (Eq. 2)	**0.893 ± 0.006**	**0.898 ± 0.005**

Median per-complex ROC AUC, mean ± std over 10 seeds. Published values taken from Table 2 of the NeurIPS paper.

- Both graph-convolution variants are **within about one std of the paper** and clearly above no-convolution.
- Ordering matches the paper: node+edge > node average > no convolution.
- Graph convolution adds +0.04–0.05 AUC over the no-convolution model (paper: +0.07–0.09).
- Node+edge: mean AUC 0.862, median AP 0.0155, P@10 6.1 % (no-conv 4.3 %).



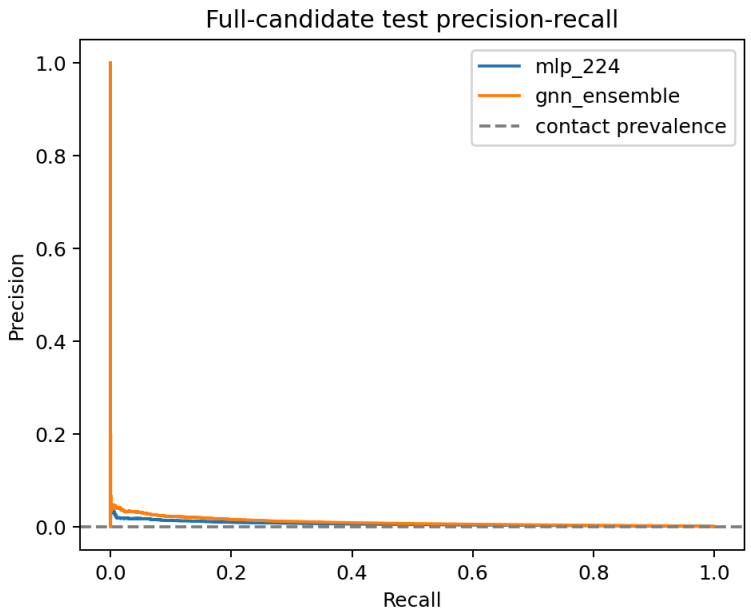
Edge-aware GNN + ensemble on 5.2 M test pairs

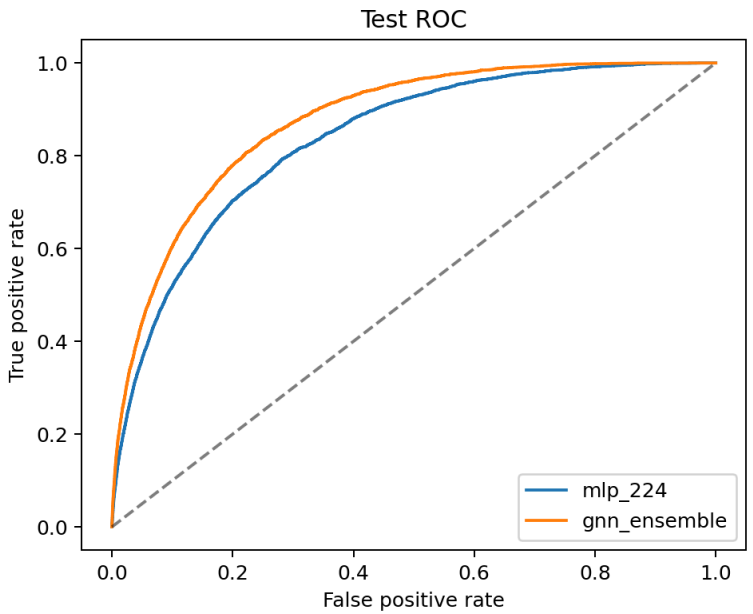
Model	Precision	Recall	F1	ROC-AUC	AP
MLP, seed 224	1.39 %	9.59 %	2.43 %	0.8354	0.00666
GNN seed 224	1.94 %	10.90 %	3.29 %	0.8646	0.00901
GNN seed 225	1.75 %	15.66 %	3.15 %	0.8626	0.00963
GNN seed 226	2.05 %	7.89 %	3.25 %	0.8673	0.00925
GNN ensemble (3 seeds)	**2.16 %**	**11.94 %**	**3.66 %**	**0.8746**	**0.01035**

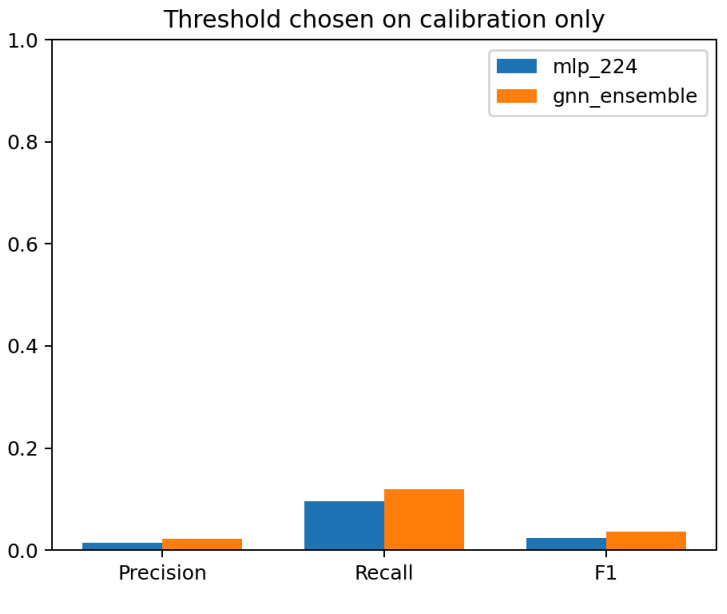
Every GNN seed beats the MLP on AUC, AP and F1; the ensemble adds **+1.24 pp F1** over the MLP. All seeds are reported, none cherry-picked.

Precision stays low. Median per-complex AUC 0.877; mean P@10 = 5.3 % (29 contacts in 550 top-10 picks).

Why AUC is high but F1 is low







Confusion (ensemble)	Count
True positives	584
False positives	26,409
False negatives	4,307

- AUC 0.875 ≠ 87 % precision: with >5 M negatives even a small false-positive rate floods the output.
- Accuracy 99.41 % *loses* to "predict nothing" (99.91 %): accuracy is the wrong headline.

What we added beyond the paper

1 · Edge-aware residual GNN

Messages weighted by distance RBF and sequence-gap features, with 3 residual layers: a deeper node+edge model than the 2-layer node-average replica.

2 · Three-seed ensemble

Equal-weight logit average, declared before test scoring; best F1 and AP of all runs.

3 · Validation-only operating point

F1 threshold calibrated on 17 held-back complexes, never on test labels.

4 · Full candidate matrix

Evaluated on all 5.2 M pairs, not on sampled negatives.

5 · Independent metric audit

4,210 recomputed checks (rank-sum AUC, grouped AP, tie-aware P@10), max discrepancy 2.2×10^{-16}.

6 · Honest uncertainty

Bootstrap CI for F1 **2.78 – 4.80 %** over 52 sequence-pair groups; prior-exposure and single-seed caveats disclosed.

Are we doing it the paper's way?

Requirement (topic row)	Our project	Status
Method: GCN	Node-average GCN + node-and-edge GCN, as in the paper	✓ met
Learn ligand and receptor residue representations	Shared-weight encoder applied to each protein	✓ met
Merge for pairwise classification	$[u+v, \lvert u-v \rvert, u \odot v] \rightarrow$ contact score	✓ met
Edges = residue interactions	20-nearest-neighbour residue graph	✓ met
Dataset: “PMC Protein Interfaces 2024”	DB5 / DB5-Plus (the paper's benchmark)	⚠ differs

Open point to confirm: the topic row lists a “PMC Protein Interfaces 2024” dataset. We used the paper's own Docking Benchmark 5 so results are comparable with the publication. If your instructor requires the PMC set, the same pipeline can be pointed at it.

What the evidence does and does not show

Supported

- Graph convolution beats the no-convolution model over 10 seeds (0.888 / 0.893 vs 0.846) and matches the paper's published range.
- Kaggle DB5-Plus run agrees: 0.875 vs 0.835 over 3 GNN seeds.
- Pipeline is leakage-controlled and its metrics were independently re-computed.

Not supported

- Reliable contact prediction: F1 only 3.7 %.
- Reliable top-10 retrieval: P@10 is only 4–6 % (node+edge 6.1 %, no-conv 4.3 %), averaged over seeds.
- A blind holdout: DB5 test structures were seen in earlier experiments; related protein families are not excluded.
- Capacity-matched comparison (MLP has fewer parameters).

Next steps: parameter-matched baseline · sequence-identity-aware splits · geometric / attention encoders (DeepInteract) · calibrated probabilities · external holdout.

REFERENCES

Sources

- Fout, Byrd, Shariat, Ben-Hur. *Protein Interface Prediction using Graph Convolutional Networks.* NeurIPS 2017. Code: github.com/fouticus/pipgcn · Data: zenodo.org/records/1127774
- Morehead et al. *Geometric Transformers for Protein Interface Contact Prediction* (DeepInteract). ICLR 2022. github.com/BioinfoMachineLearning/DeepInteract
- Morehead et al. *DIPS-Plus.* Scientific Data 2023. github.com/BioinfoMachineLearning/DIPS-Plus
- Stanford CS224W: Machine Learning with Graphs (GNN background).
- Kaggle notebook: “Protein Interface GNN – Validated GPU Training” (T4 ×2, 385.4 s, 21 Oct 2026).

> Full code, models, per-complex metrics, audit and report are in the submission ZIP.